# Momentum analysis: put it together



**antocciphysics | TEACHER KEY | CPA**

**Learning target:** Use a model to compare a deflection option and its limits.

**Suggested first pass:** 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

**01. Bring back an earlier idea (retrieve)** *Review: Unit 3, Day 10; 3 course days later.*
5 m/s west and 7 m/s south.

**02. Explain a worked example (worked)**
Internal changes cancel; total momentum change zero under negligible external impulse.

**03. Look and explain (model)**
Δv=J/M uses mass; shift=Δv t uses time. Units N·s, m/s, m.

**04. Build the solution (practice)**
Δv=.1 m/s; impulse 10 N·s.

**05. Use the idea (reason)**
ActualΔv would be smaller because mass is larger; shift smaller at same time in model.

**06. Choose a method (mixed)** *Review: Unit 3, Day 1; 12 course days later.*
Change magnitude 24 kg·m/s; force 8 N opposite initial motion.

**07. Exit check (exit)**
10 m; constant added transverse velocity/straight-line comparison.

**08. STRETCH: Apply it another way (stretch)**
Final total 9 kg·m/s; final speed 3 m/s right.

# Momentum analysis: put it together



**antocciphysics | TEACHER KEY | HONORS**

**Learning target:** Use a model to compare a deflection option and its limits.

**Suggested first pass:** 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

**01. Bring back an earlier idea (retrieve)** *Review: Unit 3, Day 10; 3 course days later.*
Other p=(12, -5) kg·m/s; magnitude≈13 kg·m/s.

**02. Explain a worked example (worked)**
Internal changes cancel; total momentum change zero under negligible external impulse.

**03. Represent and interpret (model)**
Δv=J/M uses mass; shift=Δv t uses time. Units N·s, m/s, m.

**04. Solve and justify (practice)**
Need .005 m/s and $1\times10^{7}$ N·s. Option gives .004 m/s, shift 16 km; short by 4 km in model.

**05. Analyze the reasoning (reason)**
ActualΔv would be smaller because mass is larger; shift smaller at same time in model.

**06. Choose a method (mixed)** *Review: Unit 3, Day 1; 12 course days later.*
J=-24 N·s; $F_{avg}\approx$-8 N, left.

**07. Exit check (exit)**
Unchanged: shift=(J/M) t; M×2 halvesΔv, t×2 restores shift.

**08. STRETCH: Apply it another way (stretch)**
Initial total 6; final 3; v=1 m/s. External impulse changes system momentum.